

MU XIE

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RESEARCH INTERESTS

Robotics · Robot Learning · Control & Optimization · Reinforcement Learning

EDUCATION

University of Pennsylvania, Philadelphia, PA

M.S.E., Electrical Engineering

2024.01 - 2025.05

GPA: 3.78/4.0

Relevant Coursework: Learning in Robotics, Modern Convex Optimization, Advanced Robotics, F1TENTH Autonomous Racing

Zhejiang University, Hangzhou, China

B.Eng., Electrical Engineering (Dual Degree Program with UIUC)

2019.09 - 2023.06

GPA: 3.56/4.0

Relevant Coursework: Undergrad Math Courses, Signals and Systems, Machine Learning, Numerical Analysis

University of Illinois, Urbana-Champaign

B.S., Electrical Engineering (Dual Degree with Zhejiang University)

2019.09 - 2023.06

GPA: 3.26/4.0

Relevant Coursework: Digital Signal Processing, Digital Systems, Electronic Circuits, Linear & Feedback Control Systems

PUBLICATIONS

A Hybrid Learning-to-Optimize Framework for Mixed-Integer Quadratic Programming

V.-A. Le, M. Xie, and R. Mangharam, 2025. Accepted by Learning for Dynamics and Control (L4DC), available at arXiv: <https://arxiv.org/abs/2511.19383>.

A Learning-to-Optimize Framework for Real-Time Mixed Integer Nonlinear Programming

M. Xie, V.-A. Le, and R. Mangharam, Under Submission.

RESEARCH EXPERIENCE

Research Assistant, X-Lab

2025.05 - Present

University of Pennsylvania, Philadelphia, PA

Advisor: Prof. Rahul Mangharam

- Research on safe, learning-based control and learning-to-optimize methods for autonomous systems.

Project: Learning-to-Optimize for Mixed-Integer Nonlinear Programming

- Developed the L2O-MISQP framework, integrating neural networks with a trust-region MISQP solver to accelerate solving complex MINLP problems.
- Designed an end-to-end hybrid training pipeline using differentiable optimization layers.

Project: Hybrid Learning-to-Optimize for Mixed-Integer MPC

- Proposed a hybrid learning-to-optimize framework for parametric MIQPs, targeting acceleration of mixed-integer MPC.
- Integrated neural integer prediction, a differentiable QP layer, and supervised/self-supervised losses to improve feasibility and optimality.

Project: Gaussian Process Dynamics with MPPI Control

- Learned GP dynamics from F1TENTH data and integrated uncertainty into MPPI.
- Conducted a systematic study of GP-residual dynamics in MPPI under varying levels of model mismatch, identifying conditions where learned corrections improve performance and where they cause instability or degradation.

SELECTED PROJECTS

Side Project: MPC Controller for BeamNG

2026.01 - Present

University of Pennsylvania

- Developed a BeamNG vehicle control pipeline for trajectory tracking and high-speed autonomous driving experiments.
- Implemented and evaluated LTV-MPC for tracking human-driven waypoints, achieving 80% of human-driver lap pace on the Spa track.
- Ongoing work: building Actor-Critic RL with DiffMPC and Vanilla MPPI stack in the same environment, with the main pipeline in place and final debugging in progress.

Course Project: F1TENTH Autonomous Racing

2025.01 - 2025.05

University of Pennsylvania

- Built a complete ROS 2 stack including perception, planning, and control.
- Implemented wall-following, gap-following, pure pursuit, and MPC controllers with dynamic obstacle avoidance and race-line optimization.
- Developed LiDAR-based localization and vision based vehicle detection.
- Completed two time trials and one head-to-head race on hardware. Tools: ROS 2, Python, C++.

Course Project: Quadrotor Control Paper Reproduction

2025.01 - 2025.05

University of Pennsylvania

- Reproduced nonlinear quadrotor control using cascaded attitude/position controllers.
- Analyzed robustness under parameter uncertainty and external disturbances.
- Developed simple grid-based planners and sensor-fusion routines for trajectory generation.

Course Project: LLM-Guided Navigation for F1TENTH

2025.03 - 2025.05

University of Pennsylvania

- Created an LLM-based planner converting natural-language commands to driving primitives and interacting with environment.
- Integrated symbolic reasoning with classical control for interactive autonomous-navigation tasks.
- Validated system in simulation and real F1TENTH environments.

TEACHING & INDUSTRY EXPERIENCE

Teaching Assistant, Simulation Modeling & Analysis (ESE 5030)

2025.01 - 2025.05

University of Pennsylvania, Philadelphia

- Held weekly office hours and supported approximately 80 graduate students on stochastic modeling and simulation assignments.
- Graded homework and exams, providing detailed feedback on probabilistic reasoning and modeling steps.

Internship at Fusion Array Co. Ltd

2023.07 - 2023.12

Fusion Array Co. Ltd, Hangzhou, Zhejiang

- Developed FPGA logic and test benches for an ADC chip testing system, working with high-speed interfaces and data acquisition pipelines.
- Contributed to validating chip performance under various operating conditions using Verilog HDL, Quartus, and Cadence SPB.

TECHNICAL SKILLS

Programming Languages

C/C++, PYTHON, L^AT_EX, MATLAB

Tools & Frameworks

ROS 2, PyTorch, JAX

Simulations

GYM, MUJOCO